IoT-BASED WEATHER STATION

Using ESP32, DHT22, ThingSpeak Cloud and MATLAB

IoT Mini Project Report

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1. Problem Statement

Design and implement an IoT weather monitoring system that reads temperature and humidity using a DHT22 sensor, sends the measurements to the ThingSpeak cloud platform through ESP32 Wi-Fi, displays live graphs, and triggers email alerts when predefined limits are crossed.

2. Objectives

Acquire temperature and humidity data from the DHT22 sensor using the ESP32.

Transmit the sensor readings to the ThingSpeak cloud platform using Wi-Fi.

Visualize live temperature and humidity trends using ThingSpeak charts.

Analyze the collected weather data using MATLAB.

Generate email alerts when temperature or humidity reaches the configured threshold.

3. System Architecture

The system follows a sensing-to-cloud-to-alert workflow. The DHT22 provides temperature and humidity measurements to the ESP32. The ESP32 connects to Wi-Fi and uploads the data to ThingSpeak. The cloud data is visualized through live charts and is also used by MATLAB for analysis and threshold-based alert processing.

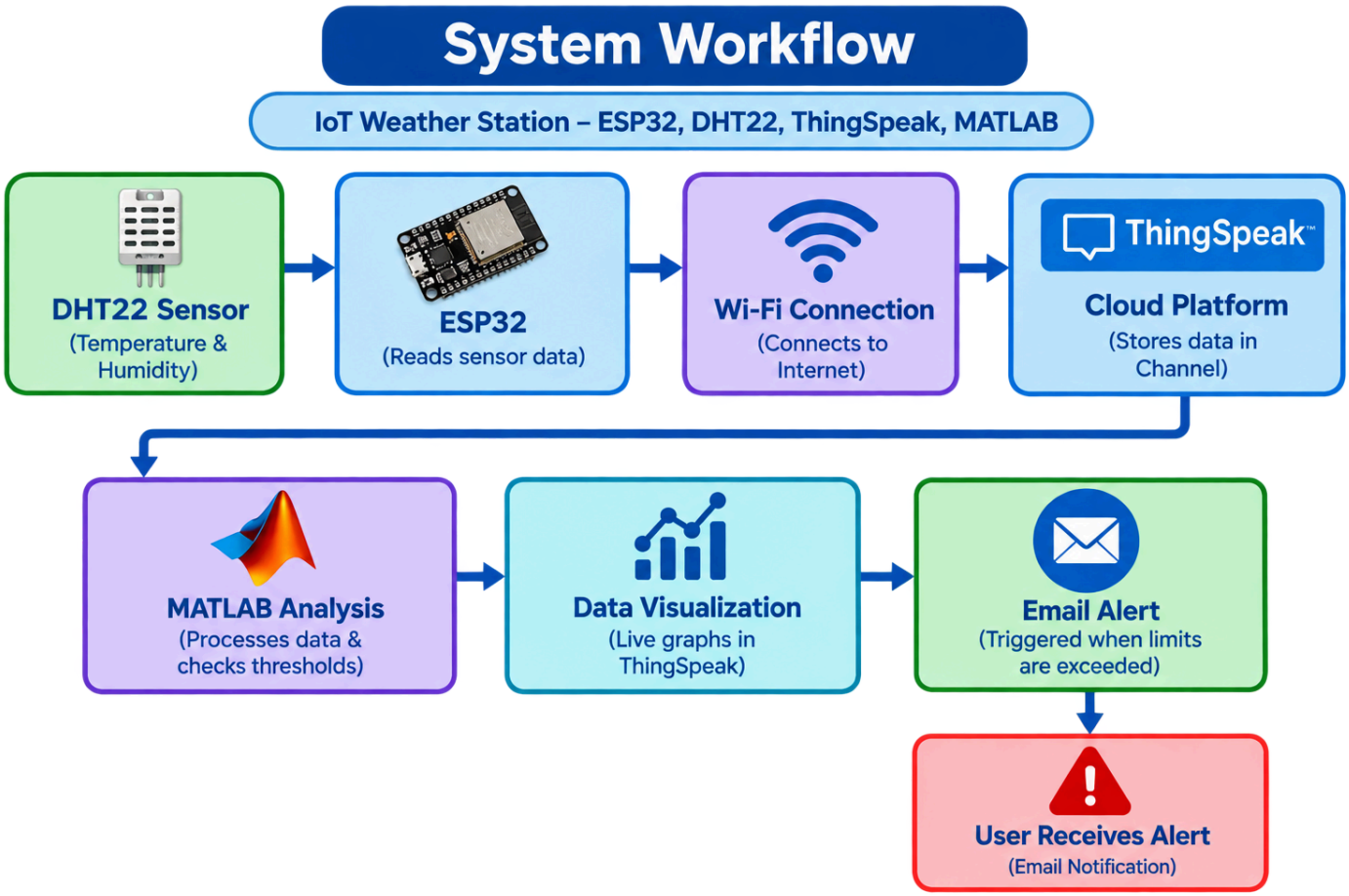


Figure 1. System architecture of the IoT weather station.

4. Hardware and Software Used

Component / Tool	Purpose	Implementation
ESP32 Development Board	Sensor acquisition and Wi-Fi connectivity	Reads DHT22 and uploads data
DHT22 Sensor	Temperature and humidity measurement	Provides environmental readings
ThingSpeak Cloud	Cloud storage and visualization	Receives Field 1 and Field 2 data
MATLAB	Data analysis and alert logic	Reads cloud values and evaluates thresholds

5. Sensor Interfacing and Firmware Logic

The DHT22 is interfaced with the ESP32 to acquire two environmental quantities: temperature and relative humidity. The firmware initializes the sensor, reads the latest values, establishes Wi-Fi connectivity, and prepares the measurements for cloud upload.

Firmware flow:

Initialize ESP32 and DHT22 sensor.
Connect the ESP32 to the configured Wi-Fi network.
Read temperature and humidity from the DHT22.
Validate the sensor readings.
Write temperature to ThingSpeak Field 1 and humidity to ThingSpeak Field 2.
Repeat the acquisition and upload cycle at the selected interval.

6. Cloud Data Upload to ThingSpeak

ThingSpeak is used as the cloud platform for receiving and visualizing the weather measurements. In the implemented channel, Field 1 is used for temperature and Field 2 is used for humidity. The live charts provide a time-based view of the uploaded values.

Field	Measured Parameter	Unit
Field 1	Temperature	°C
Field 2	Humidity	%

7. Data Visualization

The ThingSpeak dashboard was configured with separate live charts for temperature and humidity. The captured result shows the measured values changing over time and confirms that both fields are receiving data.

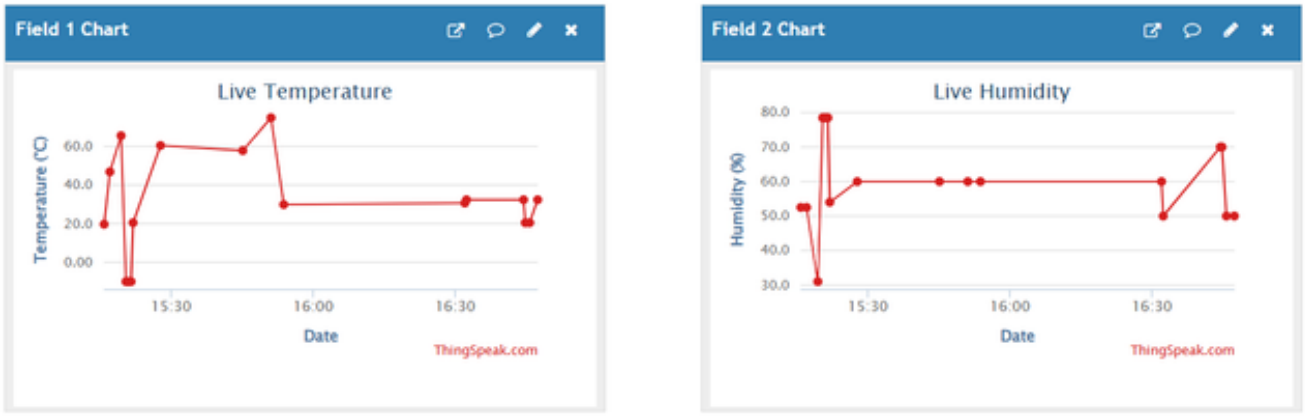


Figure 2. ThingSpeak live temperature and humidity charts.

8. MATLAB Data Analysis

MATLAB was used as an additional analysis layer for the collected weather data. The temperature and humidity datasets were analyzed separately to inspect their recorded variation. The analysis files are included with the source-code submission.

The cloud-to-analysis workflow is: ThingSpeak channel -> latest weather data -> MATLAB analysis -> threshold evaluation -> alert action.

9. Alert Mechanism

The alert mechanism compares the latest weather measurements against predefined thresholds. In the demonstrated configuration, the temperature alert limit is 30 °C and the humidity alert limit is 70 %. When a threshold is crossed or reached, ThingSpeak Alerts sends an email notification.

The MATLAB test interface also verifies the cloud value and reports whether an alert condition exists. A representative successful MATLAB run showed a temperature of 32.40 °C, which exceeded the configured 30 °C threshold and produced an accepted email request.

Condition	Observed Value	Limit	Result
High temperature	32.40 °C	30 °C	Alert triggered
High temperature	57.90 °C	30 °C	Alert triggered
High humidity	70.00 %	70 %	Alert triggered

10. Experimental Results and Alert Evidence

The following screenshots provide direct evidence of the implemented alert mechanism. The received ThingSpeak emails show that the configured thresholds generated actual alert notifications.

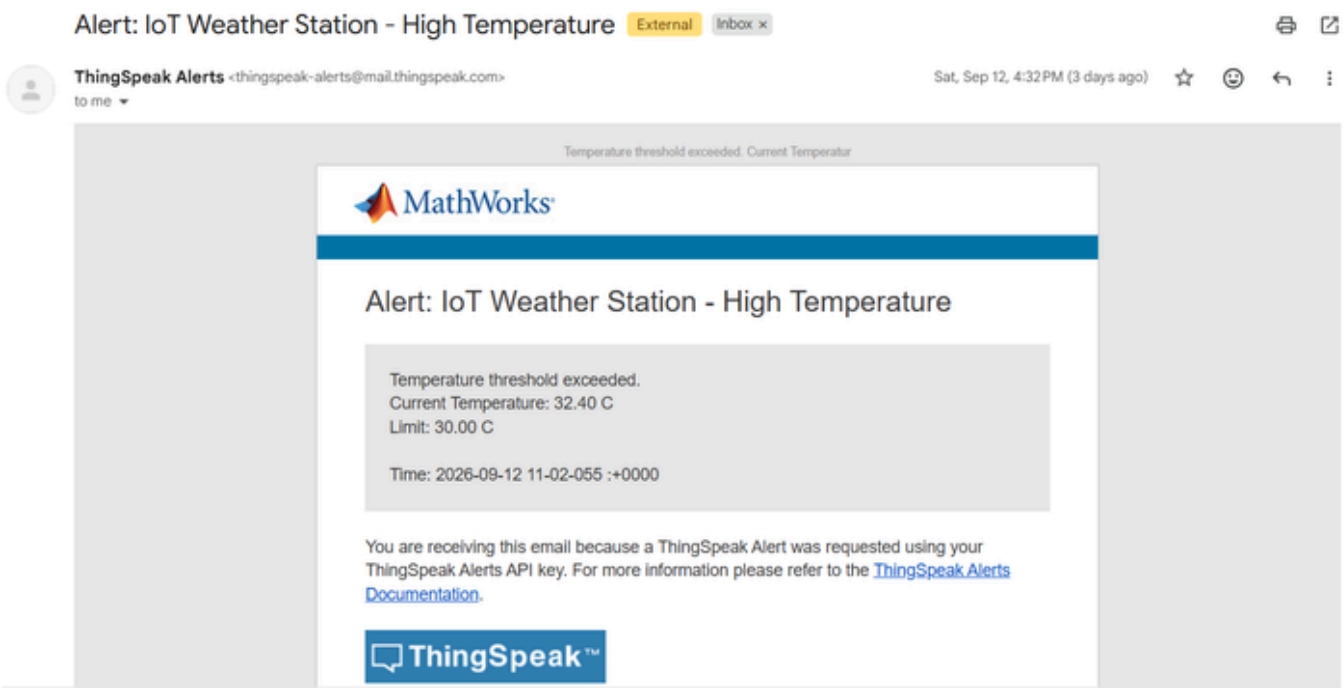


Figure 3. Received ThingSpeak email - High Temperature Alert (32.40 °C > 30 °C).

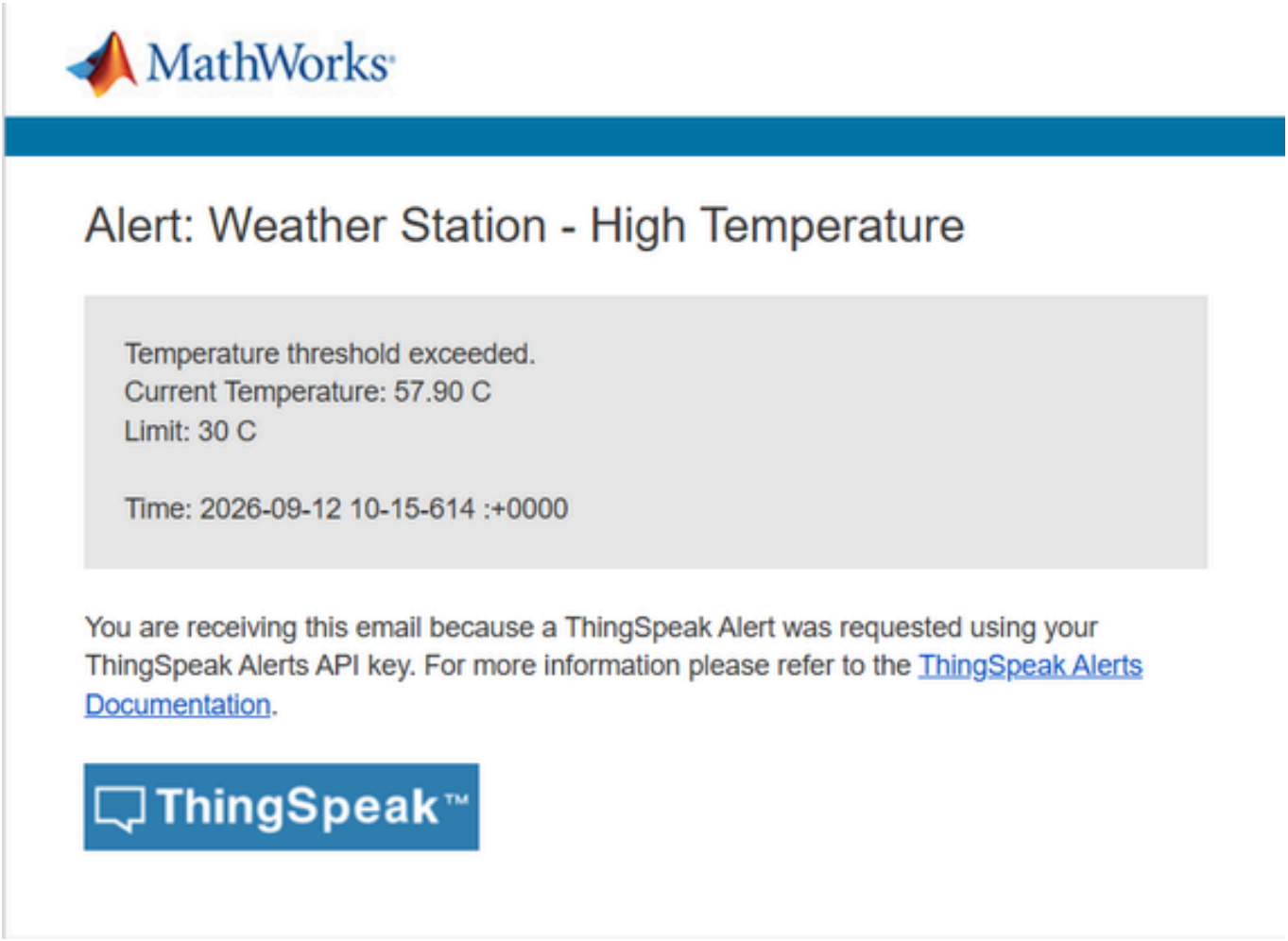


Figure 4. Received ThingSpeak email - High Temperature Alert (57.90 °C > 30 °C).

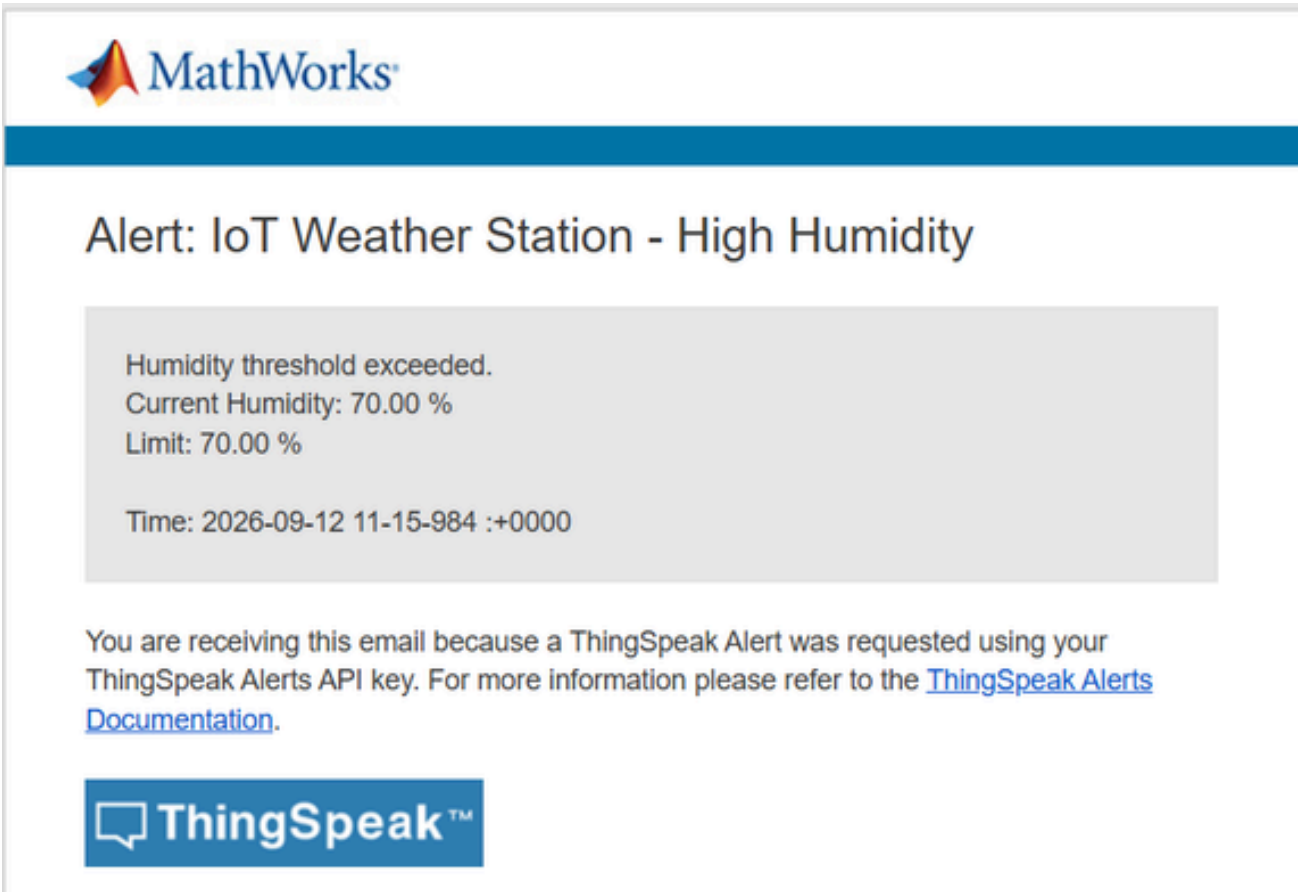


Figure 5. Received ThingSpeak email - High Humidity Alert (70.00 % at the 70 % limit).

11. Verification Against Assignment Requirements

Assignment Requirement	Implemented Work	Evidence in Report
System architecture	ESP32 + DHT22 + Wi-Fi + ThingSpeak workflow	Figure 1
Sensor interfacing	DHT22 temperature and humidity acquisition through ESP32	Section 5
Cloud data upload	Temperature to Field 1 and humidity to Field 2	Section 6 and Figure 2
Data visualization	Live ThingSpeak charts for both parameters	Figure 2
Alert mechanism	Threshold-based email alerts through ThingSpeak	Section 9 and Figures 3-5

12. Conclusion

The completed system demonstrates end-to-end IoT weather monitoring: ESP32 and DHT22 acquire temperature and humidity, thingspeak stores and visualizes the data, MATLAB supports analysis, and thingspeak Alerts generates email notifications when configured thresholds are reached or exceeded.

sketch.ino | diagram.json ● | DHT sensor library for ESPx.txt | Library Manager | libraries.txt

```ino
#include <WiFi.h>
#include <ThingSpeak.h>
#include <DHT.h>

#define DHT_PIN 4
#define DHT_TYPE DHT22

// Wokwi virtual Wi-Fi
const char* WIFI_SSID = "Wokwi-GUEST";
const char* WIFI_PASSWORD = "";

unsigned long CHANNEL_ID = 3491866;
const char* WRITE_API_KEY = "KXXR452IC5L3EGTZ";

WiFiClient client;

DHT dht(DHT_PIN, DHT_TYPE);

void setup() {

  Serial.begin(115200);

  // Start DHT22
  dht.begin();

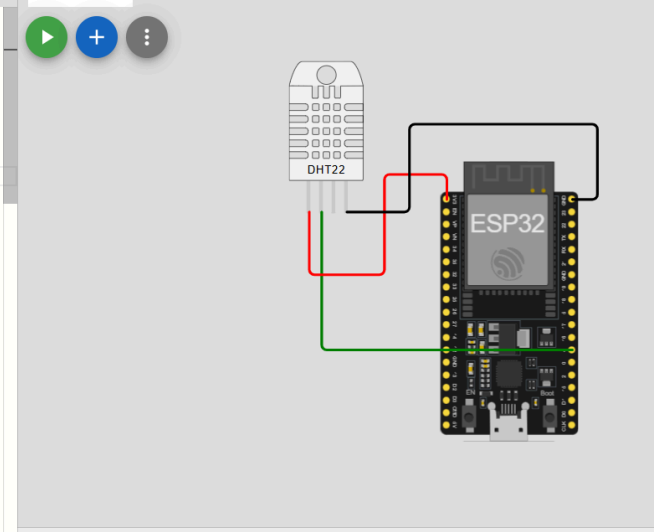
  Serial.println();
  Serial.println("================================");
  Serial.println("      IoT WEATHER STATION");
  Serial.println("================================");

  // Connect to Wokwi virtual Wi-Fi
  Serial.println("Connecting to Wi-Fi...");

  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
```

Simulation




```
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----- SENSOR DATA -----
Temperature: 32.40 °C
Humidity: 50.00 %
ThingSpeak upload successful!
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```